Response to Reviewers

Manuscript: *Agentic SABRE: An Uncertainty-Aware Neuro-Symbolic Multi-Agent Framework for Adaptive Ransomware Detection*

Expert Systems With Applications — Major Revision

April 30, 2026

We thank the Associate Editor and all four reviewers for their thorough and constructive evaluations. The comments have led to substantive improvements in the manuscript's clarity, technical rigour, and positioning. We have addressed every point raised. The changes are highlighted in **bold blue** in the revised manuscript. For conciseness, we state in each response exactly where in the revised paper the change appears.

A brief summary of the main additions:

- A new "*What Makes This Different from Ensemble-Based Approaches*" subsection in the Introduction, distinguishing SABRE's cooperative agentic design from conventional ensemble voting.
- A 7-column comparison table in Related Work summarising how prior systems differ from SABRE across key design dimensions.
- Full variable definitions and a notation glossary before the first score-vector equation; elimination of a duplicate $\sigma_{\max}$ formula; relocation of the triage-objective $J(\tau, \kappa)$ to the Policy Analysis section to remove a forward reference spanning 15 equations.
- Explicit threshold-ordering constraints ($\tau \leq \tau_{\text{high}}$, $\kappa_{\text{low}} \leq \kappa$) added as a numbered equation.
- New discussion paragraphs on: AUC= 1.0 generalisation risk; the accuracy/AUC discrepancy of the Behavioural Agent; threshold sensitivity for $\tau = 0.999999$; cross-source fusion rationale; CTGAN label clarification; Monte Carlo Dropout vs. deep ensembles; real-world executable constraints for counterfactual costs; and a runtime/complexity subsection.
- Section-opening purpose sentences in all 13 sections.

Associate Editor

> ***Reviewer Comment:*** *The authors should explicitly mention the significant contributions of the manuscript. The novelty of the paper is not highlighted.*

Response: We agree that the novelty was insufficiently foregrounded. We have added a new subsection titled *"What Makes This Different from Ensemble-Based Approaches"* immediately after the Contributions list in Section I. This subsection explains, in concrete technical terms, the three properties that distinguish Agentic SABRE from conventional ensemble detectors: (1) structurally heterogeneous input spaces ($\mathbb{R}^{384}$ semantic embeddings versus $\mathbb{R}^{16}$ I/O telemetry), so agents are complementary by design rather than redundant; (2) the triage decision conditions on epistemic *uncertainty*, not only on posterior probabilities; and (3) the triage policy is an *executable control mechanism*—not soft label aggregation—that can formally withhold autonomous action under ambiguity.

Change in manuscript: New subsection "What Makes This Different from Ensemble-Based Approaches" added in Section I after the Contributions list.

Reviewer Comment: *The advantages and limitations of the proposed approach in relationship with similar schemes is not clear.*

Response: We have added a 7-column comparison table (Table 1 in the revised manuscript, now at the end of Section III) comparing Agentic SABRE against six representative prior systems across: multi-agent design, uncertainty estimation, triage policy, symbolic cues, explainability, concept-drift handling, and score-level augmentation. On the limitations side, every section now opens with a 1–2 sentence statement of its purpose and scope, and Section XIII (Limitations and Future Work) now explicitly opens by enumerating the three principal constraints on the current design.

Change in manuscript: Comparison table (Table 1) added at the end of Section III. Section-opening sentences added to all 13 sections. Section XIII opening expanded.

Reviewer Comment: *Please revise the structure of the paper. It is recommendable to add in each section a couple of sentences that explain which is the purpose of the section.*

Response: Done. Every section in the revised manuscript now opens with 1–2 sentences stating its purpose within the overall narrative.

Change in manuscript: Purpose sentences added to the opening of Sections II through XIII.

Reviewer #1

Weaknesses

Reviewer Comment: *Weakness 1. The proposed approach is rather ensemble-based where components are parallel or consequently connected DL models (CNN and GAN-based ones). The approach itself looks more as an agent that provides decision of the sample rather than a system of cooperative agents. Thus, the title is slightly confusing.*

Response: We appreciate this important observation and address it directly in the new "What Makes This Different" subsection (Section I). While the two CNNs are trained independently and run in parallel at inference time, the system constitutes a genuine multi-agent architecture in three respects that go beyond ensemble design: (1) Each agent specialises on a structurally distinct input modality—the Semantic Agent processes PE-level embedding vectors ($\mathbb{R}^{384}$) while the Behavioural Agent processes sliding-window I/O telemetry ($\mathbb{R}^{16}$). These are not bootstrap resamples of the same feature space. (2) The agents produce not only a posterior mean probability but also an *epistemic uncertainty* estimate via Monte Carlo Dropout; this uncertainty is the primary mechanism by which the orchestrator decides whether to act autonomously. A standard ensemble does not produce per-model uncertainty signals of this kind. (3) The Decision Agent (orchestrator) is a third trained agent that operates over the score space of the first two, constituting a hierarchical agent architecture. We have clarified this framing in the revised Introduction.

Change in manuscript: New subsection "What Makes This Different from Ensemble-Based Approaches" in Section I; agent architecture clarified throughout Section IV.

Reviewer Comment: *Weakness 2. The authors list the following contribution: "lightweight symbolic reasoning using ATT&CK-aligned API ontologies and heuristic cues" (4th item). However, the "ATT&CK-aligned API ontologies" are never referred to in the approach description except for Sections II and III. This creates misleading expectations. The reasoning is done based on heuristic threshold-based rules.*

Response: The reviewer is correct. The original wording overstated the symbolic component. We have rewritten the 4th contribution bullet to accurately describe what is implemented: ATT&CK-*inspired* heuristic cues—specifically, API call category indicators, entropy-threshold signals, and ransomware-relevant filesystem access pattern flags—that are integrated into the feature extraction and triage stages. We no longer claim a formalised ontology; instead we describe these as lightweight symbolic enrichment signals grounded in ATT&CK tactic categories.

Change in manuscript: 4th contribution bullet in Section I rewritten. The revised text reads: "We incorporate **lightweight symbolic reasoning** via ATT&CK-inspired heuristic cues—including API call category indicators, entropy-threshold signals, and ransomware-relevant filesystem access patterns—integrated into the feature extraction and triage stages to enrich agent decisions with interpretable threat-intelligence signals, forming a hybrid neuro-symbolic detection pipeline."

Reviewer Comment: *Weakness 3a. Many statements are repeated in different sections: Sec. III (the last sentence), Sec. V (the last paragraph in the beginning); Sec. V A (the second paragraph); Sec. V C (the second paragraph).*

Response: We have removed or reworked all identified repetitions. Specifically: the closing sentence of Section III ("This positions SABRE as a practical neuro-symbolic ... rather than a conceptual demonstration") has been rewritten to state, concretely, which three gaps SABRE fills, rather than restating the system's positioning claim that opens Section V. The opening of Section V now serves as the primary location for this framing. Redundant restatements in Sections V-A and V-C have been condensed.

Change in manuscript: Section III final paragraph rewritten. Redundant material in Sections V-A and V-C removed.

Reviewer Comment: *Weakness 3b. The presentation in terms of mathematical formulas is difficult to follow: (i) some formulas lack explanations to the notations used—what do $p_{beh}, p_{sem}, \hat{p}$ stand for in Formula (1)? What do s_{beh}, s_{sem} stand for in Formula (1)? How does $\hat{p}$ in Formula (2) differ from $\hat{p}$ in Formula (21)? (ii) Formula 22 duplicates Formula 3. (iii) It is strange to have a reference to Formula 25 after Formula 10.*

Response: We have addressed all three sub-points.

(i) Notation before Eq. (1) and Eq. (2). We added a prose paragraph before the score-vector equation that explicitly defines $\bar{p}_{\text{sem}} \in [0, 1]$ as the predictive mean of the Semantic Agent and $\bar{p}_{\text{beh}} \in [0, 1]$ as the predictive mean of the Behavioural Agent, both obtained via MC Dropout. The notation $\mathbf{s}$ for the score vector is introduced before it appears in any formula. The fused risk $\hat{p}$ in Eq. (2) is now explicitly described as the output of the MLP Decision Agent operating on $\mathbf{s}$; a clarifying note explains that at deployment the full score vector additionally incorporates per-agent uncertainties (introduced in Section IX), so the two uses of $\hat{p}$ are consistent—both refer to the MLP's output, with the difference being whether uncertainties are included as inputs.

(ii) Duplicate $\sigma_{\max}$ formula. The canonical definition of $\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}})$ is now given once, in Section IV-C, with label `eq:sigmamax`. The occurrence in Section IX that duplicated it has been replaced by a prose reference to that equation.

(iii) Forward reference to $J(\tau, \kappa)$. The scalarised objective $J(\tau, \kappa) = \text{bal_acc}(\tau, \kappa) - \lambda\, \text{EscRate}(\tau, \kappa)$ has been moved from Section IX to Section VI (Policy Analysis), where it is first discussed. Section IX now references Eq. (12) in Section VI, eliminating the forward reference.

Change in manuscript: Variable definitions added before score-vector formula in Section IV-B. Duplicate $\sigma_{\max}$ equation removed from Section IX. $J(\tau, \kappa)$ definition relocated from Section IX to Section VI.

> ***Reviewer Comment:*** *Weakness 4. How are thresholds τ, τ_{high} ordered? How are thresholds κ, κ_{low} ordered?*

Response: We have added an explicit ordering constraint as a numbered equation in Section V-A: $\tau \leq \tau_{\text{high}}$ and $\kappa_{\text{low}} \leq \kappa$. The prose explanation now states clearly that τ_{high} demands a higher fused risk for autonomous containment than τ does for mere escalation, and that κ_{low} imposes a tighter (lower) uncertainty ceiling than κ.

Change in manuscript: Equation with ordering constraints $\tau \leq \tau_{\text{high}}$, $\kappa_{\text{low}} \leq \kappa$ added in Section V-A with accompanying explanation.

> ***Reviewer Comment:*** *Weakness 5. Figure 5 requires additional labels to axis and dotted plots.*

Response: We acknowledge this. In the revised manuscript, the caption of Figure 5 has been expanded to explicitly name each threshold line (τ, τ_{high}, κ, κ_{low}) and their roles. The figure itself will be regenerated with annotated threshold labels before the camera-ready version, as the figure generation scripts are maintained in the accompanying codebase.

Change in manuscript: Figure 5 caption expanded to annotate all four threshold lines. Figure regeneration with embedded labels is planned for the camera-ready version.

> ***Reviewer Comment:*** *Weakness 6. Why are datasets included in the approach description? Is the approach dataset-dependent?*

Response: The approach is *not* dataset-dependent. Section VIII (Experimental Setup) now opens with an explicit statement to this effect, and the datasets subsection now clarifies that RDset and RanSMAP are used to *instantiate* the two complementary modalities (static PE semantics and runtime I/O telemetry) that the approach assumes in the abstract. The architecture, triage policy, and all algorithms are specified entirely in terms of generic inputs D_{sem} and D_{beh} in Algorithms 1–4 and are fully dataset-agnostic.

Change in manuscript: Section VIII now opens with a statement clarifying that the approach is dataset-independent and that the specific datasets are used only to instantiate the two generic complementary modalities.

Reviewer Comment: *Weakness 7. Do datasets contain the same malware samples? How are features from different datasets fused? What labels are used to construct the dataset for the GAN-based model?*

Response: We have added a dedicated paragraph titled **"Cross-source fusion rationale"** in Section VIII-A addressing all three questions. In brief: (1) RDset and RanSMAP contain *no overlapping malware samples*—RDset consists of PE file metadata from Windows executables, while RanSMAP contains runtime I/O traces recorded in instrumented hypervisors. (2) Raw features are *never* merged: each agent trains on its own dataset, and fusion occurs only at the *score level* (the two posterior mean probabilities are concatenated into $\mathbf{s}$). (3) CTGAN is conditioned on binary labels $y_i \in \{0, 1\}$ (benign / ransomware) from the training set; no label transformations are applied.

Change in manuscript: "Cross-source fusion rationale" paragraph and "CTGAN labels" paragraph added in Section VIII-A.

Reviewer Comment: *Weakness 8. What is the "predefined low risk threshold θ" (p. 9, first line, Sec. IX-B)? How is it defined? What is its value?*

Response: We have added a new subsection (Section IX-A, "Counterfactual Threshold θ and Behavioural Decision Boundary") that defines θ explicitly: it is the maximum posterior probability at which the agent's output is classified as "benign" for the purpose of counterfactual generation. We set $\theta = 0.49$, just below the standard 0.5 decision boundary, to enforce a strict label flip. The subsection also reports that the 100% flip success rate is stable across $\theta \in [0.40, 0.49]$, confirming low sensitivity to this choice.

Change in manuscript: New Section IX-A "Counterfactual Threshold θ and Behavioural Decision Boundary" added, defining $\theta = 0.49$ with sensitivity note.

Reviewer #2

Reviewer Comment: *Point 1. The manuscript should more clearly distinguish its contribution from prior ensemble-based ransomware detection systems. A concise comparison paragraph highlighting what is fundamentally new would improve positioning.*

Response: Addressed by the new "What Makes This Different from Ensemble-Based Approaches" subsection in Section I and the 7-column comparison table (Table 1) in Section III. The subsection explains the three architectural properties that make SABRE a genuine multi-agent system rather than an ensemble: heterogeneous input spaces, per-agent uncertainty estimation used as an operational control signal, and a formal executable triage policy.

Change in manuscript: New subsection in Section I; comparison Table 1 in Section III.

Reviewer Comment: *Point 2. The semantic agent achieves AUC= 1.0000 on RDset. The manuscript should include a clearer discussion on generalisation risks (family-wise split, temporal split, structural similarity bias).*

Response: We have added a paragraph titled **"Generalisation caveats for AUC= 1.0"** in Section IX immediately following the AUC result. This paragraph explicitly acknowledges: (1) that RDset does not employ a family-wise or temporal split, so structural similarity bias is possible; (2) that the perfect AUC reflects the highly structured nature of PE-level metadata in an *intra-family* evaluation setting; and (3) that the saturated semantic regime is deliberately used as a stress-test for the fusion and triage layers, not as a claim of general dominance. Generalisation to temporally displaced or family-disjoint corpora is identified as future work.

Change in manuscript: "Generalisation caveats for AUC= 1.0" paragraph added in Section IX.

> ***Reviewer Comment:*** *Point 3. The behavioural agent reports high accuracy but relatively modest AUC. This discrepancy should be more clearly explained, particularly regarding class imbalance, threshold choice, and probability calibration.*

Response: We have expanded the Behavioural Agent result paragraph in Section IX with a detailed explanation of this discrepancy. In brief: the high accuracy is driven by the class prior (most samples are benign, so a low-probability classifier is accurate by default); AUC measures rank separability regardless of prior, and the modest AUC reflects the genuine overlap between malicious and benign I/O distributions in a polymorphic telemetry stream. SMOTE rebalancing improves training stability but does not resolve the distributional overlap. Crucially, the Behavioural Agent's role is framed as a *modulator of* $\sigma_{\max}$ rather than a standalone detector: its diffuse posteriors raise uncertainty for borderline cases, routing them to human analysts rather than enabling overconfident autonomous containment.

Change in manuscript: Expanded explanation of the accuracy/AUC discrepancy added in Section IX following the Behavioural Agent result.

> ***Reviewer Comment:*** *Point 4. A short discussion under Section IV comparing Monte Carlo Dropout with alternative uncertainty approaches (deep ensembles, calibration stability under distribution shift) would strengthen deployment credibility.*

Response: We have added a paragraph titled **"Choice of uncertainty estimator"** in Section IV-C (Uncertainty Aggregation). This paragraph explains why MC Dropout was chosen over deep ensembles and Bayesian neural networks: no architectural changes, lower inference cost (T forward passes versus M independently trained models), and compatibility with the existing dropout regularisation. The paragraph also honestly acknowledges that deep ensembles can produce better-calibrated uncertainty under strong distribution shift, and notes this as a limitation addressed in Section XIII.

Change in manuscript: "Choice of uncertainty estimator" paragraph added in Section IV-C.

> ***Reviewer Comment:*** *Point 5. The safety-optimal threshold* $\tau = 0.999999$ *appears extremely strict. The manuscript should clarify whether this value is dataset-specific and include a brief sensitivity or stability analysis.*

Response: We have added a paragraph titled **"Threshold sensitivity and dataset specificity"** in Section IX. This paragraph makes three points: (1) τ is dataset-specific—the near-unity value reflects the bimodal ($\{0, 1\}$-clustering) nature of the fused probabilities on RDset,

and a lower τ would be appropriate for datasets with more dispersed posteriors; (2) performance is robust to the exact value: both the safety-optimal and low-escalation policies yield near-identical accuracy (0.99999 on RDset, 0.9406 vs. 0.9405 on RanSMAP), showing headline metrics are stable; (3) the triage geometry is inherently stable near τ because the boundary neighbourhood in $(\hat{p}, \sigma_{\max})$ space is sparsely populated (argued in Section VI).

Change in manuscript: "Threshold sensitivity and dataset specificity" paragraph added in Section IX.

> ***Reviewer Comment:*** *Point 6. The adversarial cost analysis is strong. However, counterfactual norms operate in continuous space. A short clarification of how these costs translate to real executable constraints (PE format restrictions, runtime feasibility) would further enhance practical cybersecurity relevance.*

Response: We have added a paragraph titled **"Translation of counterfactual costs to real-world executable constraints"** in Section X, immediately before the Failure Modes discussion. For the Behavioural Agent, we explain that reducing LBA/GPA variance requires altering the ransomware's I/O access pattern, which conflicts directly with its encryption strategy (ATT&CK T1486: Data Encrypted for Impact), creating a functional trade-off. For the Semantic Agent, we explain that perturbing PE structural features requires maintaining PE header validity (Magic bytes, SizeOfCode consistency), bounding the feasible deformation to a subset of the continuous neighbourhood $\mathcal{N}_\epsilon(\mathbf{z})$. The reported $\mathbb{E}[\mathcal{C}_{\text{sem}}] = 3.27$ is therefore a *lower bound*; the true adversarial cost under executable constraints is higher.

Change in manuscript: "Translation of counterfactual costs to real-world executable constraints" paragraph added in Section X before the Failure Modes subsection.

Reviewer #4

> ***Reviewer Comment:*** *Point 1. The claimed innovation is somewhat overstated. The authors are encouraged to clarify the substantive technical advances over existing uncertainty-aware detection, multi-agent detection, and neuro-symbolic approaches.*

Response: We have restructured the Introduction to foreground the three substantive novelties more precisely. The new "What Makes This Different" subsection explains that: prior uncertainty-aware detectors use uncertainty as a *diagnostic output* (Kabuye et al. [**?**]), whereas SABRE uses it as an *operational control signal* that formally constrains autonomous action; prior multi-agent detectors fuse votes or feature concatenations (Soltani et al., 2023), whereas SABRE fuses heterogeneous modalities at the score level under joint uncertainty governance; and prior neuro-symbolic systems embed rules at the architecture level, whereas SABRE embeds them as interpretable triage thresholds that can be updated without model retraining. The comparison Table 1 quantifies these distinctions across six reference systems.

Change in manuscript: New subsection in Section I; comparison Table 1 in Section III; contributions list in Section I revised for precision.

Reviewer Comment: *Point 2. The dataset setting is evidently heterogeneous. The rationale for cross-source fusion, the comparability of these datasets, and the real-world deployment scenario should be further clarified.*

Response: The "Cross-source fusion rationale" paragraph in Section VIII-A directly addresses this. In brief: RDset and RanSMAP are not comparable in feature space—they are complementary observation planes of the same threat category (static PE artefacts vs. runtime I/O telemetry). Fusion at the raw feature level would be meaningless; fusion at the score level is motivated by the realistic SOC scenario in which a static sandbox scanner and a runtime endpoint monitor report simultaneously, and a security orchestrator must act on both signals. This mirrors real-world deployments where SIEM platforms ingest alerts from heterogeneous sensors. The Experimental Setup section now opens with this scenario framing.

Change in manuscript: "Cross-source fusion rationale" paragraph in Section VIII-A; Section VIII opening paragraph revised to frame the real-world deployment scenario.

Reviewer Comment: *Point 3. The threshold strategy relies heavily on parameters τ, κ, τ_{high}, κ_{low}. The parameter selection process, sensitivity analysis, and cross-dataset stability are insufficient.*

Response: We have addressed this in three ways. First, the policy objective $J(\tau, \kappa)$ is now defined in Section VI (Policy Analysis) with a full description of the optimisation procedure (grid search over the non-smooth landscape with $\lambda = 0.5$). Second, the "Threshold sensitivity and dataset specificity" paragraph in Section IX explains why $\tau = 0.999999$ is dataset-specific and demonstrates stability: both safety-optimal and low-escalation policies achieve near-identical accuracy across both datasets, confirming robustness to small perturbations in the threshold pair. Third, Section VI explains the geometric stability argument: the boundary neighbourhood in $(\hat{p}, \sigma_{\max})$ space is sparsely populated due to the monotonic fusion structure, meaning small threshold shifts move the boundary without reclassifying many samples. We acknowledge that a rigorous cross-dataset stability evaluation (e.g. on held-out ransomware families) remains an important future direction.

Change in manuscript: $J(\tau, \kappa)$ definition moved to Section VI; threshold sensitivity paragraph added in Section IX; geometric stability argument in Section VI.

Reviewer Comment: *Point 4. The manuscript repeatedly emphasises calibration, interpretability, and robustness. However, the current evidence is mainly based on qualitative analysis or limited case examples. More systematic statistical tests and comparative experiments are recommended.*

Response: We acknowledge this limitation. The existing evidence is structured across three complementary quantitative channels: (i) reliability diagrams confirming $\mathbb{P}(y = 1 \mid \hat{p} = q) \approx q$ (Figure 3); (ii) ablation tables (Tables VIII and IX) that quantify the impact of removing each component; (iii) $N = 50$ counterfactual experiments with distributional statistics (Tables VII and III). We agree that formal statistical tests (e.g. McNemar's test on the confusion matrices of the full model versus each ablated variant, or bootstrap confidence intervals on AUC differences) would further strengthen validity claims. We have noted this explicitly as a future-work item in Section XIII and have added language clarifying the nature of the evidence (empirical rather than hypothesis-test-level) at the start of Section IX.

Change in manuscript: Section IX opening paragraph clarified regarding the nature of evidence; Section XIII updated to list formal statistical validation as a future direction.

> ***Reviewer Comment:*** *Point 5. The adversarial robustness analysis is mainly based on counterfactual perturbation norms and theoretical discussion. Standard adversarial attack experiments are lacking.*

Response: We agree that standard adversarial attack experiments (e.g. FGSM or PGD on the neural agents) would strengthen robustness claims. Executing such attacks on the behavioural agent requires a differentiable model of the I/O simulator, and on the semantic agent requires a generative model of PE binaries that satisfies loader constraints—both are substantial engineering efforts outside the scope of the current paper. The counterfactual analysis serves as a principled surrogate: it computes the *minimal-norm perturbation to cross the decision boundary*, which is equivalent to the white-box ℓ_2-norm adversarial attack under a continuous relaxation. The "Translation to real-world executable constraints" paragraph (Section X) now grounds these norms in concrete PE and I/O constraints. We have added a sentence in Section XIII acknowledging standard adversarial attack evaluation as a priority for future work.

Change in manuscript: Section X adversarial cost paragraph expanded with real-world grounding. Section XIII updated to identify standard adversarial evaluation as future work.

Reviewer #5

> ***Reviewer Comment:*** *The experimental results section would benefit from including confidence intervals or variance metrics to further support the uncertainty-aware claims.*

Response: Monte Carlo Dropout itself produces per-sample uncertainty estimates σ_{sem} and σ_{beh} that can be interpreted as predictive standard deviations; these are summarised in Table VII for the counterfactual experiments. For aggregate performance metrics (AUC, accuracy), formal bootstrap confidence intervals require repeated held-out evaluation which is computationally intensive at the scale of RanSMAP (11,820 execution traces). We have added a clarifying note in Section IX stating that the MC Dropout uncertainties constitute the uncertainty evidence at the per-sample level, and that bootstrap CI computation for aggregate metrics is planned for future work. We also note that the ablation comparisons in Tables VIII and IX provide a form of model-level variance assessment across architecturally distinct configurations.

Change in manuscript: Clarifying note on confidence intervals added in Section IX. Section XIII updated with formal CI computation as a future direction.

> ***Reviewer Comment:*** *A brief discussion on computational complexity and runtime performance would improve the practical relevance of the work.*

Response: We have substantially expanded the subsection titled **"Computational Complexity and Runtime Performance"** in Section XII (Discussion) to include a formal Big-O analysis of every pipeline component, structured in six paragraphs and four numbered equations. Specifically:

- **CNN agent training** (Eq. (55)): $\mathcal{O}(E \cdot N \cdot L \cdot K \cdot C^2)$ per agent, linear in both sample count N and epoch count E, with explicit constants $L = 3$, $K = 3$, $C = 64$, and modality-specific input dimensions ($C_0 = 384$ semantic; $C_0 = 16$ behavioural).
- **MC Dropout score generation**: $\mathcal{O}(2TNLKC^2)$ for both agents combined, with $T = 30$ stochastic passes; empirical wall-clock: $\approx$ 60–150 ms per agent on CPU.
- **CTGAN augmentation** (Eq. (56)): $\mathcal{O}(E_{\text{CTGAN}} \cdot N_s \cdot [\sum_i g_i g_{i+1} + \sum_j d_j d_{j+1}])$; the score matrix is $N_s \times 2$—orders of magnitude smaller than the raw feature matrices—so 200 epochs fit in $\approx$ 5–10 min on CPU.
- **Decision Agent MLP** (Eq. (57)): $\mathcal{O}(E_{\text{MLP}} \cdot N_{\text{aug}} \cdot h^2)$, negligible relative to CTGAN given the small input dimension ($d = 4$).
- **Bayesian TPE optimisation** (Eq. (58)): $\mathcal{O}(T_{\text{TPE}} \cdot (n_{\text{obs}} \log n_{\text{obs}} + N_{\text{val}}))$, converging within minutes.
- **Per-sample inference**: dominated by MC Dropout at $\mathcal{O}(T \cdot L \cdot K \cdot C^2)$; total wall-clock $\approx$ 0.15–0.35 s CPU, $<$ 10 ms GPU.
- **Total training complexity** (Eq. (59)): a four-term closed-form expression covering all components, revealing that each CNN agent and the triage thresholds can be updated independently without retraining the rest of the pipeline.

Change in manuscript: "Computational Complexity and Runtime Performance" subsection in Section XII substantially expanded with formal Big-O analysis of all pipeline components (four new numbered equations: (55), (56), (57), (58), (59)) and six structured paragraphs.

> ***Reviewer Comment:*** *The related work section may be strengthened by including a brief comparison table summarising key differences between existing approaches and the proposed framework.*

Response: Done. Table 1 in the revised Section III compares Agentic SABRE against six representative systems across seven design dimensions: multi-agent architecture, uncertainty estimation, operational triage policy, symbolic cues, explainability, concept-drift handling, and score-level augmentation. The table uses a three-level encoding (✓ = present; ∘ = partial; × = absent) for each dimension.

Change in manuscript: Comparison table (Table 1) added at the end of Section III.

> ***Reviewer Comment:*** *The manuscript is generally well-written; however, a few grammatical inconsistencies and typographical errors should be carefully reviewed.*

Response: The entire manuscript has been re-read and edited for grammatical consistency and typographical accuracy. Key corrections include: uniform use of the en-dash vs. hyphen in compound adjectives; removal of doubled words and inconsistent capitalisation in section headings; and standardised notation for all mathematical symbols (e.g. consistent use of $\bar{p}$ vs. $\hat{p}$ and σ vs. $\sigma_{\max}$ across sections).

Change in manuscript: Manuscript-wide grammar and typographical review completed; corrections applied throughout.

Closing Remark

We believe the revised manuscript is substantially stronger in its technical precision, structural clarity, and contextual positioning. We are grateful to all reviewers for the depth of their engagement. We remain available to address any remaining concerns the editor or reviewers may have.

The Authors